

Artificial Variables Technique and Big-M Method

1 Artificial Variables Technique

In the simplex method, we need an **initial basic feasible solution (BFS)** to start the calculations. For a constraint of the form $\leq$, a slack variable usually provides a starting basic variable. However, when a constraint contains $\geq$ or $=$, the required identity matrix may not be obtained directly. To overcome this difficulty, we introduce some temporary variables called **artificial variables**.

Artificial Variable: An **artificial variable** is a temporary variable introduced into a constraint only for the purpose of obtaining an initial basic feasible solution.

Artificial variables do not have any physical meaning in the original linear programming problem. Therefore, they must be removed from the basis during the simplex iterations. **Artificial variables are temporary variables only.**

When Do We Introduce Artificial Variables?

There are three important cases.

Case 1: Constraint of Type $\leq$: Consider $ax + by \leq c$. We introduce a **slack variable** s : $ax + by + s = c$. Thus, $\boxed{\leq \Rightarrow +\text{Slack Variable}}$. No artificial variable is normally required.

Case 2: Constraint of Type $\geq$: Consider $ax + by \geq c$. First subtract a **surplus variable** s : $ax + by - s = c$. This does not give a suitable basic variable. Hence, we also introduce an artificial variable A : $ax + by - s + A = c$. Therefore,

$$\boxed{\geq \Rightarrow -\text{Surplus Variable} + \text{Artificial Variable}}$$

Case 3: Constraint of Type $=$: Consider $ax + by = c$. There is neither a slack variable nor a surplus variable. Therefore, an artificial variable A is introduced: $ax + by + A = c$. Hence, $\boxed{= \Rightarrow +\text{Artificial Variable}}$

Big-M Method

The **Big-M Method** is used for solving linear programming problems in which artificial variables are required. **The main purpose of the method is to remove the artificial variables from the basis.** Here, M represents a very large positive number.

Artificial variables do not belong to the original L.P.P. Therefore, a very large penalty is assigned to them in the objective function.

For a **maximization problem**, we assign the coefficient $-M$ to every artificial variable.

Thus, $\boxed{\text{For maximization: artificial-variable coefficient} = -M}$ For example, if Maximize $Z =$

$3x_1 + 5x_2$ and A_1, A_2 are artificial variables, then the modified objective function is $Z = 3x_1 + 5x_2 - MA_1 - MA_2$.

For a **minimization problem**, we normally assign $+M$ to each artificial variable. Thus,

Problem	Penalty
Maximization	$-M$
Minimization	$+M$

Why is a Large Penalty Used?

Since artificial variables are not part of the original problem, we want their values to become zero. The large penalty M forces the artificial variables to leave the basis during the simplex iterations. Once an artificial variable leaves the basis, it is normally not allowed to enter the basis again.

Steps of the Big-M Method

The procedure of the Big-M Method is as follows.

Step 1: Make sure that the right-hand side of every constraint is non-negative.

Step 2: Convert each inequality into an equation.

Step 3: For a $\leq$ constraint, add a slack variable.

Step 4: For a $\geq$ constraint, subtract a surplus variable and add an artificial variable.

Step 5: For an equality constraint, add an artificial variable.

Step 6: Modify the objective function by assigning a large penalty to each artificial variable. For maximization use $-MA_i$, while for minimization use $+MA_i$.

Step 7: Form the initial simplex table.

Step 8: Calculate the net evaluation values $\Delta_j = c_j - Z_j$, $Z_j = \sum_i c_{B_i} a_{ij} = c_B^T Y_j$.

Step 9: Select the entering variable. For a maximization problem using $\Delta_j = c_j - Z_j$, the column having the largest positive Δ_j is selected.

Step 10: Apply the minimum ratio test: $\frac{x_{B_i}}{y_{ij}}$, $y_{ij} > 0$. The row having the smallest positive ratio gives the leaving variable.

Step 11: Perform the pivot operation.

Step 12: Repeat the simplex calculations until the optimality condition is satisfied.

Note: At the end of the Big-M Method, all artificial variables must be removed from the basis. If an artificial variable remains in the final basis with a positive value, then the original L.P.P. has no feasible solution. Thus,

Positive artificial variable in final solution $\Rightarrow$ L.P.P. is infeasible.

Example 1: Apply the Big-M Method to solve the following L.P.P.:

$$\text{Maximize } Z = 2x_1 + 4x_2$$

subject to

$$2x_1 + x_2 \leq 18,$$

$$3x_1 + 2x_2 \geq 30,$$

$$x_1 + 2x_2 = 26,$$

$$x_1, x_2 \geq 0.$$

Solution: The problem is a maximization problem.

Step 1: Convert all the constraints into equations.

From the first constraint, $2x_1 + x_2 \leq 18$. Since it is a $\leq$ constraint, introduce a slack variable x_3 : $2x_1 + x_2 + x_3 = 18$. From the second constraint, $3x_1 + 2x_2 \geq 30$. Since it is a $\geq$ constraint, subtract a surplus variable x_4 and add an artificial variable A_1 : $3x_1 + 2x_2 - x_4 + A_1 = 30$. From the third constraint, $x_1 + 2x_2 = 26$. Since it is an equality constraint, introduce an artificial variable A_2 : $x_1 + 2x_2 + A_2 = 26$.

Therefore, the constraints become

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 18, \\ 3x_1 + 2x_2 - x_4 + A_1 &= 30, \\ x_1 + 2x_2 + A_2 &= 26. \end{aligned}$$

Objective function becomes $Z = 2x_1 + 4x_2 + 0x_3 + 0x_4 - MA_1 - MA_2$.

Step 2: Initial Basic Feasible Solution

Initially, take the non-basic variables as $x_1 = x_2 = x_4 = 0$. Then, $x_3 = 18$, $A_1 = 30$, $A_2 = 26$. Therefore, the starting basic variables are x_3 , A_1 , A_2 , and the initial basic feasible solution is $x_3 = 18$, $A_1 = 30$, $A_2 = 26$.

First Simplex Table

B	c_B	c_j	2	4	0	0	$-M$	$-M$	Min. ratio x_B/x_2 , $y_{i2} > 0$
		x_B	x_1	x_2	x_3	x_4	A_1	A_2	
x_3	0	18	2	1	1	0	0	0	18
A_1	$-M$	30	3	2	0	-1	1	0	15
A_2	$-M$	26	1	2	0	0	0	1	$\frac{26}{2} = 13$ (min)
$Z = c_B^T x_B = -56M$	Δ_j		$2 + 4M$	$4 + 4M$ $\uparrow$	0	$-M$	0	0	x_2 enters

Step 3: Optimality Test

We use $\Delta_j = c_j - Z_j$, $Z_j = c_B^T Y_j$, where $c_B^T = (0, -M, -M)$.

For x_1 , $\Delta_1 = 2 - (0, -M, -M) \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = 2 - (-4M) = 2 + 4M$. Therefore, $\Delta_1 = 2 + 4M$.

For x_2 , $\Delta_2 = 4 - (0, -M, -M) \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = 4 - (-4M) = 4 + 4M$. Therefore, $\Delta_2 = 4 + 4M$.

For x_3 , $\Delta_3 = 0$. For x_4 , $\Delta_4 = 0 - (0, -M, -M) \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} = 0 - M = -M$. Hence, $\Delta_4 = -M$. Also, $\Delta_{A_1} = 0$, $\Delta_{A_2} = 0$. Since $\Delta_1 > 0$ and $\Delta_2 > 0$, the current basic feasible solution with $Z = -56M$ is not optimal.

Step 4: Find the Entering and Leaving Variables

Since $4 + 4M > 2 + 4M$, x_2 enters the basis.

Minimum Ratio Test: Consider the x_2 column. The positive ratios are $\frac{18}{1} = 18$, $\frac{30}{2} = 15$, $\frac{26}{2} = 13$. Therefore, $\min\{18, 15, 13\} = 13$. Hence, x_2 enters the basis and A_2 leaves the basis. The pivot element is 2. Hence the new basic variables are x_3, A_1, x_2 .

Step 5: Row Operations for the First Pivot

We have $R_1 = [18, 2, 1, 1, 0, 0, 0]$, $R_2 = [30, 3, 2, 0, -1, 1, 0]$, $R_3 = [26, 1, 2, 0, 0, 0, 1]$.

Since the pivot element is 2 in the third row, first divide the third row by 2:

$$R_3^{\text{new}} = \frac{R_3}{2} = \frac{1}{2}[26, 1, 2, 0, 0, 0, 1] = \left[13, \frac{1}{2}, 1, 0, 0, 0, \frac{1}{2}\right].$$

For the first row, $R_1^{\text{new}} = R_1 - R_3^{\text{new}} = [18, 2, 1, 1, 0, 0, 0] - \left[13, \frac{1}{2}, 1, 0, 0, 0, \frac{1}{2}\right] = \left[5, \frac{3}{2}, 0, 1, 0, 0, -\frac{1}{2}\right]$.

For the second row, $R_2^{\text{new}} = R_2 - 2R_3^{\text{new}} = [30, 3, 2, 0, -1, 1, 0] - 2 \left[13, \frac{1}{2}, 1, 0, 0, 0, \frac{1}{2}\right] = [4, 2, 0, 0, -1, 1, -1]$.

Second Simplex Table

B	c_B	c_j	2	4	0	0	$-M$	$-M$	Min. ratio $x_B/x_1, y_{i1} > 0$
		x_B	x_1	x_2	x_3	x_4	A_1	A_2	
x_3	0	5	$\frac{3}{2}$	0	1	0	0	$-\frac{1}{2}$	$\frac{10}{3}$
A_1	$-M$	4	2	0	0	-1	1	-1	2 (min)
x_2	4	13	$\frac{1}{2}$	1	0	0	0	$\frac{1}{2}$	26
$Z = c_B^T x_B = 52 - 4M$	Δ_j		$2M$ $\uparrow$	0	0	$-M$	0	$-2M - 2$	x_1 enters

Step 6: Optimality Test for the Second Simplex Table

The basic-cost vector is $c_B^T = (0, -M, 4)$. For x_1 , $\Delta_1 = 2 - (0, -M, 4) \begin{pmatrix} \frac{3}{2} \\ 2 \\ \frac{1}{2} \end{pmatrix} = 2 - (-2M + 2) = 2M > 0$.

Similarly, $\Delta_2 = \Delta_3 = 0$, $\Delta_4 = -M$. Thus, the solution $Z = 52 - 4M$ is not optimal. Since $\Delta_1 = 2M$ is the largest positive net evaluation, x_1 is the entering variable.

For the leaving variable, the minimum ratios are $\frac{5}{3/2} = \frac{10}{3}$, $\frac{4}{2} = 2$, $\frac{13}{1/2} = 26$. Therefore, $\min \left\{ \frac{10}{3}, 2, 26 \right\} = 2$. Hence, A_1 leaves the basis. The new basic variables are x_3, x_1, x_2 .

Step 7: Row Operations for the Second Pivot

We have, $R_1 = [5, \frac{3}{2}, 0, 1, 0, 0, -\frac{1}{2}]$, $R_2 = [4, 2, 0, 0, -1, 1, -1]$, $R_3 = [13, \frac{1}{2}, 1, 0, 0, 0, \frac{1}{2}]$. Since the pivot element is 2 in the second row, first divide the second row by 2: $R_2^{\text{new}} = \frac{R_2}{2} = \frac{1}{2}[4, 2, 0, 0, -1, 1, -1] = [2, 1, 0, 0, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}]$.

For the first row, $R_1^{\text{new}} = R_1 - \frac{3}{2}R_2^{\text{new}} = [5, \frac{3}{2}, 0, 1, 0, 0, -\frac{1}{2}] - \frac{3}{2}[2, 1, 0, 0, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}] = [2, 0, 0, 1, \frac{3}{4}, -\frac{3}{4}, \frac{1}{4}]$.

For the third row, $R_3^{\text{new}} = R_3 - \frac{1}{2}R_2^{\text{new}} = [13, \frac{1}{2}, 1, 0, 0, 0, \frac{1}{2}] - \frac{1}{2}[2, 1, 0, 0, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}] = [12, 0, 1, 0, \frac{1}{4}, -\frac{1}{4}, \frac{3}{4}]$.

Third Simplex Table (Final Table)

B	c_B	c_j	2	4	0	0	$-M$	$-M$	Result
		x_B	x_1	x_2	x_3	x_4	A_1	A_2	
x_3	0	2	0	0	1	$\frac{3}{4}$	$-\frac{3}{4}$	$\frac{1}{4}$	
x_1	2	2	1	0	0	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	
x_2	4	12	0	1	0	$\frac{1}{4}$	$-\frac{1}{4}$	$\frac{3}{4}$	
$Z = c_B^T x_B = 52$		Δ_j	0	0	0	0	$-M$	$-M - 2$	$\Delta_j \leq 0$ Optimal

Step 8: Final Optimality Test and Solution

For the final table, the basic-cost vector is $c_B^T = (0, 2, 4)$. The net evaluations are $\Delta_1 = 0$, $\Delta_2 = 0$, $\Delta_3 = 0$, $\Delta_4 = 0$, and for the artificial-variable columns, $\Delta_{A_1} = -M$, $\Delta_{A_2} = -M - 2$. Therefore, $\Delta_j \leq 0$ for every column. Hence, the current basic feasible solution is optimal.

From the final table, $x_1 = 2$, $x_2 = 12$, $x_3 = 2$, $x_4 = 0$, and both artificial variables are non-basic, so $A_1 = A_2 = 0$. Thus, $x_1 = 2, x_2 = 12$. The maximum value of the objective function is $Z_{\text{max}} = 2(2) + 4(12) = 52$.

Example 2: Apply Big-M method to solve the L.P.P.

$$\text{Maximize} \quad Z = -x_1 - x_2$$

subject to

$$\begin{aligned}
3x_1 + 2x_2 &\geq 30, \\
-2x_1 + 3x_2 &\leq -30, \\
x_1 + x_2 &\leq 5, \\
x_1, x_2 &\geq 0.
\end{aligned}$$

2 Revised Simplex Method

Introduction

The **Revised Simplex Method** is an improved computational form of the ordinary simplex method used for solving linear programming problems.

The ordinary simplex method works with the complete simplex table at every iteration. However, a large part of the information contained in the simplex table is not required for deciding the entering and leaving variables. The main idea of the revised simplex method is:

Compute and store only the information required at the current iteration instead of computing the entire simplex table.

Hence, the revised simplex method is particularly useful for solving large linear programming problems with the help of computers.

Why is it Called the Revised Simplex Method?

The method follows essentially the same principle as the ordinary simplex method.

The difference lies mainly in the **computational procedure**.

In the ordinary simplex method, the complete simplex tableau is updated after every pivot.

In the revised simplex method, we mainly work with B , B^{-1} , x_B , $Y_j = B^{-1}a_j$, and the reduced costs $\Delta_j = c_j - Z_j$. Therefore, unnecessary calculations involving the complete tableau are avoided.

Standard Forms

There are two commonly used standard forms in the revised simplex method.

Standard Form I

Standard Form I is used when an initial basic feasible solution can be obtained using only slack or surplus variables and **artificial variables are not required**.

Standard Form II

Standard Form II is used when **artificial variables are required** to obtain an initial basis.

In such cases, the artificial variables may be handled by the **Two-Phase Method** or another suitable artificial-variable technique.

Standard Form I of a Revised Simplex Method

Consider the linear programming problem

$$\text{Maximize} \quad Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

subject to

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &\leq b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &\leq b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &\leq b_m, \end{aligned}$$

with $x_j \geq 0$, $j = 1, 2, \dots, n$.

1. After introducing slack variables, the problem can be written in matrix form as

$$Ax = b, \quad x \geq 0. \text{ The objective function is } Z = c^T x. \text{ Here, } A = \begin{bmatrix} a_{11} & a_{12} & \cdots \\ a_{21} & a_{22} & \cdots \\ \vdots & \vdots & \ddots \\ a_{m1} & a_{m2} & \cdots \end{bmatrix},$$

$$\text{and } b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

2. **Basis Matrix:** Suppose the current basic variables are $x_{B_1}, x_{B_2}, \dots, x_{B_m}$.

The columns of A corresponding to these basic variables form the **basis matrix**

$$B = \begin{bmatrix} | & | & & | \\ \beta_1 & \beta_2 & \cdots & \beta_m \\ | & | & & | \end{bmatrix}. \text{ The matrix } B \text{ is an } m \times m \text{ nonsingular matrix. Therefore,}$$

its inverse B^{-1} exists. The revised simplex method mainly uses B^{-1} instead of repeatedly writing the entire simplex tableau.

3. **Basic Solution:** Since $Bx_B = b$, the vector of basic variables is $\boxed{x_B = B^{-1}b}$. Thus, once B^{-1} is known, the current basic solution can be obtained directly. For the solution to be a basic feasible solution, we require $x_B = B^{-1}b \geq 0$.

4. **Value of the Objective Function:** Let $c_B = \begin{bmatrix} c_{B_1} & c_{B_2} & \cdots & c_{B_m} \end{bmatrix}$ be the coefficients of the basic variables in the objective function. Then the current value of the objective function is $Z = c_B x_B$. Since $x_B = B^{-1}b$, we may also write $Z = c_B B^{-1}b$.

5. **Computation for a Non-Basic Variable:** Let a_j be a column of A corresponding to a non-basic variable x_j . Compute $Y_j = B^{-1}a_j$. If $Y_j = \begin{bmatrix} y_{1j} \\ y_{2j} \\ \vdots \\ y_{mj} \end{bmatrix}$, then the

corresponding Z_j value is $Z_j = c_B Y_j$. Therefore, the net evaluation or reduced cost is $\Delta_j = c_j - Z_j$. Hence, $\Delta_j = c_j - c_B B^{-1}a_j$.

This is one of the most important formulas in the revised simplex method.

6. **Optimality Test:** For a **maximization problem**, using $\Delta_j = c_j - Z_j$, the current basic feasible solution is optimal if $\Delta_j \leq 0$ for every non-basic variable. If some $\Delta_j > 0$, then the current solution can still be improved.

7. **Entering Variable:** If the solution is not optimal, select the non-basic variable having the largest positive value of Δ_j . That is, $\Delta_k = \max\{\Delta_j : \Delta_j > 0\}$. Then x_k enters the basis. The corresponding column required for the ratio test is $Y_k = B^{-1}a_k$.

8. **Leaving Variable: Minimum Ratio Test**

Let $Y_k = \begin{bmatrix} y_{1k} \\ y_{2k} \\ \vdots \\ y_{mk} \end{bmatrix}$ be the column corresponding to the entering variable. The leaving

variable is determined by $\theta = \min_{y_{ik} > 0} \left\{ \frac{(x_B)_i}{y_{ik}} \right\}$. The basic variable corresponding to the minimum positive ratio leaves the basis. Thus,

Leaving variable = variable corresponding to the minimum positive ratio.

The element at the intersection of the entering column and leaving row is called the **pivot element** or **key element**.

9. **Unbounded Solution:** If $\Delta_k > 0$ for an entering variable but all the components of $Y_k = B^{-1}a_k$ satisfy $y_{ik} \leq 0$, then no positive ratio can be formed. In this case, the objective function can increase without bound. Hence, the L.P.P. has an **unbounded solution**.

Computational Procedure of the Revised Simplex Method

The following steps are used to solve an L.P.P. by the revised simplex method.

- Step 1:** If the problem is a minimization problem, convert it into an appropriate maximization form, if required.
- Step 2:** Convert the constraints into equations by introducing slack, surplus, and artificial variables as necessary.
- Step 3:** Ensure that $b_i \geq 0$.
- Step 4:** Identify an initial basic feasible solution.
- Step 5:** Form the initial basis matrix B .
- Step 6:** Find B^{-1} .
- Step 7:** Compute the current basic solution $\boxed{x_B = B^{-1}b}$.
- Step 8:** Compute the objective-function value $\boxed{Z = c_B x_B}$.
- Step 9:** For every non-basic variable x_j , calculate $Y_j = B^{-1}a_j$. Then calculate $Z_j = c_B Y_j$ and $\boxed{\Delta_j = c_j - Z_j}$.
- Step 10:** Apply the optimality test.
For a maximization problem, if $\Delta_j \leq 0$ for every non-basic variable, then the current solution is optimal.
- Step 11:** If the solution is not optimal, choose the variable corresponding to the largest positive Δ_j as the entering variable.
- Step 12:** For the entering variable x_k , calculate $Y_k = B^{-1}a_k$.
- Step 13:** Apply the minimum ratio test $\boxed{\theta = \min_{y_{ik} > 0} \left\{ \frac{(x_B)_i}{y_{ik}} \right\}}$. The variable corresponding to the minimum ratio leaves the basis.
- Step 14:** Replace the leaving column of the basis by the entering column and obtain the new basis matrix.
- Step 15:** Compute or update the new B^{-1} and repeat the procedure.
- Step 16:** Continue the iterations until $\boxed{\Delta_j \leq 0}$ for all non-basic variables.

Ordinary Simplex Method vs. Revised Simplex Method

Ordinary Simplex Method	Revised Simplex Method
The complete simplex tableau is constructed.	The complete simplex tableau is generally not required.
Most entries of the tableau are recalculated at every iteration.	Only the quantities required for the current iteration are calculated.
Requires more storage for large problems.	Requires less storage.
May involve many unnecessary arithmetic calculations.	Avoids many unnecessary calculations.
Convenient for small problems and hand calculations.	More suitable for large problems and computer calculations.
Works directly with the simplex table.	Mainly works with B^{-1} , x_B , Y_j and reduced costs.

Advantages of the Revised Simplex Method

The main advantages of the revised simplex method over the ordinary simplex method are:

1. **Less computation:** It does not calculate every entry of the complete simplex tableau at every iteration.
2. **Less computer memory:** Only the basis matrix, its inverse or related information, and the required vectors are stored.
3. **Efficient for large problems:** When a linear programming problem contains a large number of variables, the complete simplex table may become very large. The revised simplex method avoids storing the whole table.
4. **Suitable for computer implementation:** The method mainly uses matrix and vector operations, which can be performed efficiently on computers.
5. **Same principle as the simplex method:** The rules for selecting entering and leaving variables remain essentially the same as in the ordinary simplex method.
6. **Only necessary information is computed:**

Example 1: Solve the following L.P.P. by the Revised Simplex Method:

$$\text{Maximize} \quad Z = x_1 + 2x_2$$

subject to

$$\begin{aligned}x_1 + x_2 &\leq 3, \\x_1 + 2x_2 &\leq 5, \\3x_1 + x_2 &\leq 6, \\x_1, x_2 &\geq 0.\end{aligned}$$

Solution: Step 1: Convert the L.P.P. into standard form.

Introduce slack variables x_3, x_4, x_5 . Then

$$\begin{aligned}x_1 + x_2 + x_3 &= 3, \\x_1 + 2x_2 + x_4 &= 5, \\3x_1 + x_2 + x_5 &= 6.\end{aligned}$$

The objective function is $Z = x_1 + 2x_2$. Therefore, $A = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 \\ 3 & 1 & 0 & 0 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix}$,

$c = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \end{bmatrix}$. Since the columns corresponding to the slack variables x_3, x_4, x_5 form an identity matrix, an initial basic feasible solution is obtained without using artificial variables. Hence, the problem is in **Revised Simplex Standard Form I**.

Step 2: Find the initial basic feasible solution. Take x_3, x_4, x_5 as the initial basic

variables. Therefore, $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$. Hence, $B^{-1} = I_3$. The initial basic solution is

$$x_B = B^{-1}b. \text{ Therefore, } x_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix}. \text{ Thus, } \boxed{x_3 = 3, x_4 = 5, x_5 = 6}.$$

The non-basic variables are $x_1 = x_2 = 0$. Also, $c_B = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$. Therefore, the initial value of the objective function is $Z = c_B x_B = 0$.

Step 3: Test the initial solution for optimality.

For every non-basic variable x_j , compute $Y_j = B^{-1}a_j$ and $\Delta_j = c_j - c_B Y_j$. Since $B^{-1} = I_3$

and $c_B = (0, 0, 0)$, for x_1 , $Y_1 = B^{-1}a_1 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$, and hence $\Delta_1 = c_1 - c_B Y_1 = 1 - 0 = 1$.

Similarly, for x_2 , $Y_2 = B^{-1}a_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, so $\Delta_2 = c_2 - c_B Y_2 = 2 - 0 = 2$. Thus,

$\boxed{\Delta_1 = 1, \Delta_2 = 2}$. For a maximization problem, the current solution is optimal only when $\Delta_j \leq 0$ for all non-basic variables. Since $\Delta_1 > 0$, $\Delta_2 > 0$, the current solution is not optimal.

First Revised Simplex Table

Basis	c_B	B^{-1}			x_B	$Y_2 = B^{-1}a_2$	Min. Ratio $\frac{x_B}{Y_2}$
		1	2	3			
x_3	0	1	0	0	3	1	$\frac{3}{1} = 3$
x_4	0	0	1	0	5	$\boxed{2}$	$\boxed{\frac{5}{2}}$
x_5	0	0	0	1	6	1	$\frac{6}{1} = 6$

Step 4: Find the incoming and outgoing variables. Since $\Delta_2 = \max\{\Delta_1, \Delta_2\} = 2$, we choose x_2 as the entering variable. For the entering variable x_2 , $Y_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$. Apply the minimum ratio test: $\frac{3}{1} = 3$, $\frac{5}{2} = \frac{5}{2}$, $\frac{6}{1} = 6$. Therefore, $\min\{3, \frac{5}{2}, 6\} = \frac{5}{2}$. The minimum ratio corresponds to x_4 . Hence, $\boxed{x_4 \text{ leaves the basis.}}$ Thus, $\boxed{x_2 \text{ enters and } x_4 \text{ leaves.}}$ The pivot element is 2. Therefore, the new basic variables are x_3, x_2, x_5 .

Step 5: Construct the new basis matrix.

The columns corresponding to the new basic variables x_3, x_2, x_5 are $a_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $a_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, $a_5 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Hence, the new basis matrix is $B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. Its inverse is $B^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix}$.

Step 6: Find the improved basic feasible solution.

We have $x_B = B^{-1}b$. Therefore, $x_B = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 - \frac{5}{2} \\ \frac{5}{2} \\ 6 - \frac{5}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{5}{2} \\ \frac{7}{2} \end{bmatrix}$.

Thus, $x_3 = \frac{1}{2}$, $x_2 = \frac{5}{2}$, $x_5 = \frac{7}{2}$. The non-basic variables are now $x_1 = x_4 = 0$. Therefore, $x_1 = 0$, $x_2 = \frac{5}{2}$. The value of the objective function is $Z = x_1 + 2x_2 = 0 + 2\left(\frac{5}{2}\right) = 5$.

Hence, $Z = 5$.

Second Revised Simplex Table

Basis	c_B	B^{-1}			x_B
		1	2	3	
x_3	0	1	$-\frac{1}{2}$	0	$\frac{1}{2}$
x_2	2	0	$\frac{1}{2}$	0	$\frac{5}{2}$
x_5	0	0	$-\frac{1}{2}$	1	$\frac{7}{2}$

Step 7: Test the improved solution for optimality.

For the new basis, $c_B = \begin{bmatrix} 0 & 2 & 0 \end{bmatrix}$. The non-basic variables are x_1, x_4 . We calculate their reduced costs.

For x_1 : $a_1 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$. Thus, $Y_1 = B^{-1}a_1 = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{5}{2} \end{bmatrix}$.

Now, $Z_1 = c_B Y_1 = \begin{bmatrix} 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{5}{2} \end{bmatrix} = 1$. Hence, $\Delta_1 = c_1 - Z_1 = 1 - 1 = 0$.

For x_4 : $a_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. Therefore, $Y_4 = B^{-1}a_4 = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$.

Thus, $Z_4 = c_B Y_4 = \begin{bmatrix} 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} = 1$. Hence, $\Delta_4 = c_4 - Z_4 = 0 - 1 = -1$.

The final reduced costs can be summarized as follows:

Non-basic Variable	Z_j	$\Delta_j = c_j - Z_j$
x_1	1	0
x_4	1	-1

we have $\Delta_j \leq 0$ for all non-basic variables. Therefore, the current solution is optimal.

Optimal Solution:

Hence, one optimal basic feasible solution is $x_1 = 0$, $x_2 = \frac{5}{2}$. The maximum value of the objective function is $Z_{\max} = 5$.

Example 2: Solve the following L.P.P. by the Revised Simplex Method:

$$\text{Maximize} \quad Z = 2x_1 + x_2$$

subject to

$$3x_1 + 4x_2 \leq 6,$$

$$6x_1 + x_2 \leq 3,$$

$$x_1, x_2 \geq 0.$$

Solution:

Step 1: Convert the L.P.P. into standard form.

Introduce slack variables x_3 and x_4 . Then

$$3x_1 + 4x_2 + x_3 = 6,$$

$$6x_1 + x_2 + x_4 = 3.$$

The objective function is $Z = 2x_1 + x_2$. Therefore, $A = \begin{bmatrix} 3 & 4 & 1 & 0 \\ 6 & 1 & 0 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$, and $c = \begin{bmatrix} 2 & 1 & 0 & 0 \end{bmatrix}$. Since the columns corresponding to the slack variables x_3 and x_4 form the identity matrix, an initial basic feasible solution is obtained without artificial variables. Hence, the problem is in **Revised Simplex Standard Form I**.

Step 2: Find the initial basic feasible solution. Take x_3, x_4 as the initial basic variables. Thus, $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$, $B^{-1} = I_2$. The initial basic solution is $x_B = B^{-1}b = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$. Therefore, $x_3 = 6$, $x_4 = 3$. The non-basic variables are $x_1 = x_2 = 0$. Also,

$c_B = \begin{bmatrix} 0 & 0 \end{bmatrix}$. Hence, $Z = c_B x_B = 0$.

Step 3: Test the initial solution for optimality.

For every non-basic variable x_j , $Y_j = B^{-1}a_j$ and $\Delta_j = c_j - c_B Y_j$. For x_1 , $Y_1 = B^{-1}a_1 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$. Therefore, $\Delta_1 = 2 - (0, 0) \begin{bmatrix} 3 \\ 6 \end{bmatrix} = 2$. For x_2 , $Y_2 = B^{-1}a_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$. Hence, $\Delta_2 = 1 - (0, 0) \begin{bmatrix} 4 \\ 1 \end{bmatrix} = 1$. Thus, $\Delta_1 = 2$, $\Delta_2 = 1$. Since positive Δ_j values occur, the initial solution is not optimal. Since $\Delta_1 = \max\{2, 1\} = 2$, we have x_1 enters the basis.

First Revised Simplex Table

Basis	c_B	B^{-1}		x_B	$Y_1 = B^{-1}a_1$	Min. Ratio
		1	2			$\frac{x_B}{Y_1}$
x_3	0	1	0	6	3	$\frac{6}{3} = 2$
x_4	0	0	1	3	6	$\frac{3}{6} = \frac{1}{2}$

Step 4: Find the leaving variable.

For the entering variable x_1 , $Y_1 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$. The minimum ratio test gives $\frac{6}{3} = 2$, $\frac{3}{6} = \frac{1}{2}$.

Hence, $\min \{2, \frac{1}{2}\} = \frac{1}{2}$. Therefore, x_4 leaves the basis. Thus, x_1 enters and x_4 leaves. The pivot element is 6 The new basic variables are x_3, x_1 .

Step 5: Construct the new basis matrix.

The columns corresponding to x_3 and x_1 are $a_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $a_1 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$. Hence, $B =$

$\begin{bmatrix} 1 & 3 \\ 0 & 6 \end{bmatrix}$. Therefore, $B^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{6} \end{bmatrix}$. The new basic solution is $x_B = B^{-1}b$. Thus,

$$x_B = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 6 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 - \frac{3}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{9}{2} \\ \frac{1}{2} \end{bmatrix}. \quad \text{Therefore, } x_3 = \frac{9}{2}, x_1 = \frac{1}{2}. \text{ The non-basic}$$

variables are $x_2 = x_4 = 0$. The objective-function value is $Z = 2 \left(\frac{1}{2}\right) = 1$.

Step 6: Test the improved solution for optimality.

For the present basis, $c_B = \begin{bmatrix} 0 & 2 \end{bmatrix}$.

The non-basic variables are x_2 and x_4 .

For x_2 , $a_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$. Hence, $Y_2 = B^{-1}a_2 = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{7}{2} \\ \frac{1}{6} \end{bmatrix}$.

Therefore, $Z_2 = c_B Y_2 = (0, 2) \begin{bmatrix} \frac{7}{2} \\ \frac{1}{6} \end{bmatrix} = \frac{1}{3}$. Thus, $\Delta_2 = c_2 - Z_2 = 1 - \frac{1}{3} = \frac{2}{3}$.

For x_4 , $Y_4 = B^{-1}a_4 = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{6} \end{bmatrix}$. Hence, $Z_4 = (0, 2) \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{6} \end{bmatrix} = \frac{1}{3}$. Therefore, $\Delta_4 = 0 - \frac{1}{3} = -\frac{1}{3}$. Since $\Delta_2 = \frac{2}{3} > 0$, the solution is not optimal.

Therefore, x_2 enters the basis.

Second Revised Simplex Table

Basis	c_B	B^{-1}		x_B	$Y_2 = B^{-1}a_2$	Min. Ratio $\frac{x_B}{Y_2}$
		1	2			
x_3	0	1	$-\frac{1}{2}$	$\frac{9}{2}$	$\frac{7}{2}$	$\frac{9/2}{7/2} = \frac{9}{7}$
x_1	2	0	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1/2}{1/6} = 3$

Step 7: Find the leaving variable.

The positive ratios are $\frac{9/2}{7/2} = \frac{9}{7}$, and $\frac{1/2}{1/6} = 3$. Therefore, $\min\{\frac{9}{7}, 3\} = \frac{9}{7}$. Hence, x_3 leaves the basis. Thus, x_2 enters and x_3 leaves. The pivot element is $\frac{7}{2}$. Therefore, the new basic variables are x_2, x_1 .

Step 8: Construct the new basis matrix.

The columns corresponding to the new basic variables x_2 and x_1 are $a_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $a_1 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$. Therefore, $B = \begin{bmatrix} 4 & 3 \\ 1 & 6 \end{bmatrix}$. Since $|B| = 24 - 3 = 21$, we obtain $B^{-1} = \frac{1}{21} \begin{bmatrix} 6 & -3 \\ -1 & 4 \end{bmatrix} =$

$$\begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ -\frac{1}{21} & \frac{4}{21} \end{bmatrix}.$$

Step 9: Find the new basic solution.

We have $x_B = B^{-1}b$. Therefore, $x_B = \begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ -\frac{1}{21} & \frac{4}{21} \end{bmatrix} \begin{bmatrix} 6 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{12}{7} - \frac{3}{7} \\ -\frac{6}{21} + \frac{12}{21} \end{bmatrix} = \begin{bmatrix} \frac{9}{7} \\ \frac{2}{7} \end{bmatrix}$. Since the basis is (x_2, x_1) , $x_2 = \frac{9}{7}$, $x_1 = \frac{2}{7}$. The non-basic variables are $x_3 = x_4 = 0$. The value of the objective function is $Z = 2x_1 + x_2 = 2\left(\frac{2}{7}\right) + \frac{9}{7} = \frac{4}{7} + \frac{9}{7} = \frac{13}{7}$. Thus, $Z = \frac{13}{7}$.

Third Revised Simplex Table

Basis	c_B	B^{-1}		x_B
		1	2	
x_2	1	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{9}{7}$
x_1	2	$-\frac{1}{21}$	$\frac{4}{21}$	$\frac{2}{7}$

Step 10: Final optimality test.

For the final basis, $c_B = \begin{bmatrix} 1 & 2 \end{bmatrix}$. The non-basic variables are x_3 and x_4 . For x_3 , $a_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Hence, $Y_3 = B^{-1}a_3 = \begin{bmatrix} \frac{2}{7} \\ 1 \\ -\frac{1}{21} \end{bmatrix}$. Therefore, $Z_3 = c_B Y_3 = (1, 2) \begin{bmatrix} \frac{2}{7} \\ 1 \\ -\frac{1}{21} \end{bmatrix} = \frac{2}{7} - \frac{2}{21} = \frac{4}{21}$.

Thus, $\Delta_3 = c_3 - Z_3 = -\frac{4}{21}$. For x_4 , $a_4 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Therefore, $Y_4 = B^{-1}a_4 = \begin{bmatrix} -\frac{1}{7} \\ 4 \\ \frac{1}{21} \end{bmatrix}$.

Hence, $Z_4 = c_B Y_4 = (1, 2) \begin{bmatrix} -\frac{1}{7} \\ 4 \\ \frac{1}{21} \end{bmatrix} = -\frac{1}{7} + \frac{8}{21} = \frac{5}{21}$. Therefore, $\Delta_4 = c_4 - Z_4 = -\frac{5}{21}$.

The final reduced costs are $\Delta_3 = -\frac{4}{21}$, $\Delta_4 = -\frac{5}{21}$. Since $\Delta_j \leq 0$ for all non-basic variables, the present basic feasible solution is optimal.

Optimal Solution: Hence, $x_1 = \frac{2}{7}$, $x_2 = \frac{9}{7}$ and the maximum value of the objective function is $Z_{\max} = \frac{13}{7}$.